

gilmore-lie-groups: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ TAB_001	26	0.877	$\begin{tabular}{t}{ l l l l l }\hline & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{tabular}$	$\begin{array}{c ccc c} & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{array}$	$\begin{array}{c ccc c} & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{array}$
gilmore-lie-groups_ TAB_002	30	0.669	$\begin{tabular}{t}{ l l l l l l }\hline & (I) & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{tabular}$	$\begin{array}{c ccccc c} & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{array}$	$\begin{array}{c ccccc c} & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{array}$
gilmore-lie-groups_ TAB_003	62	0.837	$\begin{tabular}{t}{ l l }\hline & \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{tabular}$	$\begin{array}{c c } \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{array}$	$\begin{array}{c c } \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{array}$
gilmore-lie-groups_ TAB_004	62	1.000	$\begin{tabular}{t}{ l l l }\hline & \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{tabular}$	$\begin{array}{c cc } \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{array}$	$\begin{array}{c cc } \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{array}$

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gilmore-lie-groups_ TAB_011	172	■ 1.000	<pre> \begin{tabular}[t]{l l l l l l} \hline \multirow{b}{3}{*}{\((i \ \phi)^{j} \ f_{j}\)} & & & & & \\ \multicolumn{4}{ c }{Representation} \ \hline & & & & \\ \frac{1}{2} & 1 & & & 2 \\ & 3 & & & 5 \\ \hline f_0 & \cos(\phi/2) & 1 & \frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2}) & 1 \\ (i \ \phi)^1 f_1 & 2i \sin(\phi/2) & i \sin(\phi) & \frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2}) & \frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi) \\ (i \ \phi)^2 f_2 & & \cos(\phi) - 1 & -\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2}) & -\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi) \\ (i \ \phi)^3 f_3 & & & -i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2}) & -\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi) \\ (i \ \phi)^4 f_4 & & & & \frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi) \\ \hline \hline </pre>	(not rendered)	
gilmore-lie-groups_ TAB_012	175	■ 0.995	<pre> \begin{tabular}[t]{l l l l l l} \hline \cos ^{2}(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ 3 & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{3}{4} & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 1 & \pm 3 & 3^{+1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 2 & \pm 1 & 2^{-1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 2 & 2^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 1 & 1 \\ 0 & \frac{\pi}{2} & 0 & 0 & - \\ \hline \hline </pre>	<pre> \begin{tabular}[t]{l l l l l l} \hline \cos ^{2}(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ 3 & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{3}{4} & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 1 & \pm 3 & 3^{+1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 2 & \pm 1 & 2^{-1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 2 & 2^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 1 & 1 \\ 0 & \frac{\pi}{2} & 0 & 0 & - \\ \hline \hline </pre>	

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gilmore-lie-groups_ TAB_013	180	0.508	$\begin{tabular}{t}{ l l l l l l }\hline & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ \hline A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & " & " & " & \cdots & " & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & " & " & " & \cdots & " & \mathbf{e}_l \\ C_l & " & " & " & \cdots & " & 2\mathbf{e}_l \\ \hline \end{tabular}$	<table><tr><td></td><td>α_1</td><td>α_2</td><td>α_3</td><td></td><td>α_{l-1}</td><td>α_l</td></tr><tr><td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>...</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td><td></td></tr><tr><td>D_l</td><td>"</td><td>"</td><td>"</td><td>...</td><td>"</td><td>$\mathbf{e}_{l-1} + \mathbf{e}_l$</td></tr><tr><td>$B_l$</td><td>"</td><td>"</td><td>"</td><td>...</td><td>"</td><td>$\mathbf{e}_l$</td></tr><tr><td>C_l</td><td>"</td><td>"</td><td>"</td><td>...</td><td>"</td><td>$2\mathbf{e}_l$</td></tr></table>		α_1	α_2	α_3		α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	...	$\mathbf{e}_{l-1} - \mathbf{e}_l$		D_l	"	"	"	...	"	$\mathbf{e}_{l-1} + \mathbf{e}_l$	B_l	"	"	"	...	"	$\mathbf{e}_l$	C_l	"	"	"	...	"	$2\mathbf{e}_l$	$\begin{array}{ccccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & " & " & " & \cdots & " & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & " & " & " & \cdots & " & \mathbf{e}_l \\ C_l & " & " & " & \cdots & " & 2\mathbf{e}_l \end{array}$					
	α_1	α_2	α_3		α_{l-1}	α_l																																							
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D_l	"	"	"	...	"	$\mathbf{e}_{l-1} + \mathbf{e}_l$																																							
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C_l	"	"	"	...	"	$2\mathbf{e}_l$																																							
gilmore-lie-groups_ TAB_014	182	0.999	$\begin{tabular}{t}{ l l l l }\hline & (p) & (q) & (r) \\ \hline & \text{Root space} & & \\ \hline 3 & 3 & 2 & E_6 \\ 4 & 3 & 2 & E_7 \\ 5 & 3 & 2 & E_8 \\ \hline \end{tabular}$	<table><tr><td>p</td><td>q</td><td>r</td><td>Root space</td></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
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gilmore-lie-groups_ TAB_015	184	1.000	$\begin{tabular}{t}{ l l l l }\hline & \text{Canonical form} & \text{Boson operators} & \text{Coordinates and derivatives} \\ \hline H_i & & b_i^\dagger b_i & x^i \partial_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & & b_i^\dagger b_j & x^i \partial_j \\ \hline \end{tabular}$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																
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gilmore-lie-groups_ TAB_016	184	■ 0.960	<pre>\begin{tabular}[t]{ l l l l l } \hline \multirow[b]{2}{*}{Canonical form} & \multicolumn{2}{ c }{\((C_l)\)} & \multicolumn{2}{ c }{\((D_l)\)} \\ \hline & \hline & Boson operators & Coordinates and derivatives & Fermion operators & Coordinates and derivatives \\ \hline H_i & $b_i^\dagger b_i + \frac{1}{2}$ & $x^i \partial_i$ & $f_i^\dagger f_i + \frac{1}{2}$ & $x^i \partial_i + \frac{1}{2}$ \\ $E_{+e_i - e_j}$ & $b_i^\dagger b_j$ & $x^i \partial_j$ & $f_i^\dagger f_j$ & $x^i \partial_j$ \\ $E_{+e_i + e_j}$ & $b_i^\dagger b_j^\dagger$ & $x^i x^j$ & $f_i^\dagger f_j^\dagger$ & $x^i x^j$ \\ $E_{-e_i - e_j}$ & $b_i b_j$ & $\partial_i \partial_j$ & $f_i f_j$ & $\partial_i \partial_j$ \\ E_{+2e_i} & $b_i^\dagger b_i^\dagger$ & $x^i x^i$ & & \\ E_{-2e_i} & $b_i b_i$ & $\partial_i \partial_i$ & & \\ \hline </pre>	(not rendered)	<table><tr><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+e_i - e_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+e_i + e_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-e_i - e_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2e_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>E_{-2e_i}</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	C_l		D_l		Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+e_i - e_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+e_i + e_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-e_i - e_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	E_{+2e_i}	$b_i^\dagger b_i^\dagger$	$x^i x^i$			E_{-2e_i}	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ TAB_017	195	■ 0.899	<pre>\begin{tabular}[t]{ l l l l l } \hline Mapping & Real form & Root space & Condition \\ \hline Block submatrix & $\mathfrak{so}(p, q)$ & D_n & $p + q = 2n$ \\ & $\mathfrak{so}(p, q)$ & B_n & $p + q = 2n + 1$ \\ & $\mathfrak{su}(p, q)$ & A_{n-1} & $p + q = n$ \\ & $\mathfrak{sp}(p, q)$ & C_n & $p + q = n$ \\ Subfield restriction & $\mathfrak{sl}(n; \mathbb{R})$ & A_{n-1} & \\ & $\mathfrak{sp}(2n; \mathbb{R})$ & C_n & \\ Field embedding & $\mathfrak{so}^*(2n)$ & D_n & \\ & $\mathfrak{su}^*(2n)$ & A_{2n-1} & \\ \hline </pre>	(not rendered)																																								

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gilmore-lie-groups_ TAB_018	195	■ 0.500	$\begin{tabular}{t l l l l l } \hline (A_1) & \sim & (B_1) & \sim & (C_1) \\ (B_1) & \sim & (C_1) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \hline \mathfrak{u}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ \hline D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ \hline B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ \hline D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{tabular}$	<table><tr><td>A_1</td><td></td><td>B_1</td><td></td><td>C_1</td></tr><tr><td>$\mathfrak{su}(2)$</td><td></td><td>$\mathfrak{so}(3)$</td><td></td><td>$\mathfrak{sp}(1) = \mathfrak{usp}(2)$</td></tr><tr><td>$\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R})$</td><td></td><td>$\mathfrak{so}(2, 1)$</td><td></td><td>$\mathfrak{sp}(2; \mathbb{R})$</td></tr><tr><td>$D_2$</td><td>=</td><td>$A_1$</td><td>+</td><td>$A_1$</td></tr><tr><td>$\mathfrak{so}(4)$</td><td>=</td><td>$\mathfrak{so}(3)$</td><td>+</td><td>$\mathfrak{so}(3)$</td></tr><tr><td>$\mathfrak{so}^*(4)$</td><td></td><td>$\mathfrak{so}(3)$</td><td>+</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>$\mathfrak{so}(3, 1)$</td><td></td><td>$\mathfrak{sl}(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(2, 2)$</td><td></td><td>$\mathfrak{so}(2, 1)$</td><td>+</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>B_2</td><td>=</td><td>C_2</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5)$</td><td></td><td>$\mathfrak{sp}(2) = \mathfrak{usp}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 1)$</td><td></td><td>$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 2)$</td><td></td><td>$\mathfrak{sp}(4; R)$</td><td></td><td></td></tr><tr><td>D_3</td><td>=</td><td>A_3</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(6)$</td><td></td><td>$\mathfrak{su}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5, 1)$</td><td></td><td>$\mathfrak{su}^*(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}^*(6)$</td><td></td><td>$\mathfrak{su}(3, 1)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 2)$</td><td></td><td>$\mathfrak{su}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 3)$</td><td></td><td>$\mathfrak{sl}(4; \mathbb{R})$</td><td></td><td></td></tr></table>	A_1		B_1		C_1	$\mathfrak{su}(2)$		$\mathfrak{so}(3)$		$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R})$		$\mathfrak{so}(2, 1)$		$\mathfrak{sp}(2; \mathbb{R})$	D_2	=	A_1	+	A_1	$\mathfrak{so}(4)$	=	$\mathfrak{so}(3)$	+	$\mathfrak{so}(3)$	$\mathfrak{so}^*(4)$		$\mathfrak{so}(3)$	+	$\mathfrak{so}(2, 1)$	$\mathfrak{so}(3, 1)$		$\mathfrak{sl}(2; \mathbb{C})$			$\mathfrak{so}(2, 2)$		$\mathfrak{so}(2, 1)$	+	$\mathfrak{so}(2, 1)$	B_2	=	C_2			$\mathfrak{so}(5)$		$\mathfrak{sp}(2) = \mathfrak{usp}(4)$			$\mathfrak{so}(4, 1)$		$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$			$\mathfrak{so}(3, 2)$		$\mathfrak{sp}(4; R)$			D_3	=	A_3			$\mathfrak{so}(6)$		$\mathfrak{su}(4)$			$\mathfrak{so}(5, 1)$		$\mathfrak{su}^*(4)$			$\mathfrak{so}^*(6)$		$\mathfrak{su}(3, 1)$			$\mathfrak{so}(4, 2)$		$\mathfrak{su}(2, 2)$			$\mathfrak{so}(3, 3)$		$\mathfrak{sl}(4; \mathbb{R})$		
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gilmore-lie-groups_ TAB_019	196	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline \multirow[b]{2}{*}{Root space} & \multirow[b]{2}{*}{Class \({}_\text{rank(character)}\)} & \multicolumn{2}{ l }{Maximal compact subgroup} \\ \hline & & Root space & Dimension \\ \hline G_2 & $G_{2(-14)}$ & G_2 & 14 \\ & $G_{2(+2)}$ & $A_1 + A_1$ & 6 \\ F_4 & $F_{4(-52)}$ & F_4 & 52 \\ & $F_{4(-20)}$ & B_4 & 36 \\ & $F_{4(+4)}$ & $C_3 + A_1$ & 24 \\ E_6 & $E_{6(-78)}$ & E_6 & 78 \\ & $E_{6(-26)}$ & F_4 & 52 \\ & $E_{6(-14)}$ & $D_5 + D_1$ & 46 \\ & $E_{6(+2)}$ & $A_5 + A_1$ & 38 \\ & $E_{6(+6)}$ & C_4 & 36 \\ E_7 & $E_{7(-133)}$ & E_7 & 133 \\ & $E_{7(-25)}$ & $E_6 + D_1$ & 79 \\ & $E_{7(-5)}$ & $D_6 + A_1$ & 69 \\ & $E_{7(+7)}$ & A_7 & 63 \\ E_8 & $E_{8(-248)}$ & E_8 & 248 \\ & $E_{8(-24)}$ & $E_7 + A_1$ & 136 \\ & $E_{8(+8)}$ & D_8 & 120 \\ \hline \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$															
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gilmore-lie-groups_ TAB_020	199	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline Quadrant & Operator & Edge \\ \hline I & J_+ & $n_2 = 0$ \\ I & J_- & $n_1 = 0$ \\ III & J_+ & $n_1 = -1$ \\ III & J_- & $n_2 = -1$ \\ \hline \hline \end{tabular}</pre>	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
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gilmore-lie-groups_ TAB_021	206	0.877	$\begin{tabular}{t l l l l l } \hline \text{Root space} \& \text{Quotient } (G^{\prime}/H) \& \text{Dimension } (P) \& \text{Rank } (P) \\ \hline (A_{p+q-1}) \& (SU(p,q)/S[U(p)\otimes U(q)]) \& 2pq \& \min(p,q) \\ (A_{n-1}) \& (SL(n;\mathbb{R})/SO(n)) \& \frac{1}{2}(n+2)(n-1) \& n-1 \\ (A_{2n-1}) \& (SU^*(2n)/USp(2n)) \& (2n+1)(n-1) \& n-1 \\ (B_{p+q}) \& (SO(p,q)/SO(p)\otimes SO(q)) \& pq \& \min(p,q) \\ (D_{p+q}) \& (SO(p,q)/SO(p)\otimes SO(q)) \& pq \& \min(p,q) \\ (D_n) \& (SO^*(2n)/U(n)) \& n(n-1) \& [n/2] \\ (C_{p+q}) \& (USp(2p,2q)/USp(2p)\otimes USp(2q)) \& 4pq \& \min(p,q) \\ (C_n) \& (Sp(2n;\mathbb{R})/U(n)) \& n(n+1) \& n \\ \hline \end{tabular}$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n						
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gilmore-lie-groups_ TAB_022	207	0.979	$\begin{tabular}{t l l l l l l } \hline \text{Root space} \& (G^{\prime}/H) \& (\operatorname{Dim} G^{\prime}) \& (\operatorname{Dim} H) \& (\operatorname{Dim} P) \& \text{Rank } (P) \\ \hline (G_2) \& (G_{2(+2)}) \& 14 \& 6 \& 8 \& 2 \\ (F_4) \& (F_{4(-20)}) \& 52 \& 36 \& 16 \& 1 \\ (F_4) \& (F_{4(+4)}) \& 52 \& 24 \& 28 \& 4 \\ (E_6) \& (E_{6(-26)}) \& 78 \& 52 \& 26 \& 2 \\ (E_6) \& (E_{6(-14)}) \& 78 \& 46 \& 32 \& 2 \\ (E_6) \& (E_{6(+2)}) \& 78 \& 38 \& 40 \& 4 \\ (E_6) \& (E_{6(+6)}) \& 78 \& 36 \& 42 \& 6 \\ (E_6) \& (E_{7(-25)}) \& 133 \& 79 \& 54 \& 3 \\ (E_6) \& (E_{7(-5)}) \& 133 \& 69 \& 64 \& 4 \\ (E_6) \& (E_{7(+7)}) \& 133 \& 63 \& 70 \& 7 \\ (E_8) \& (E_{8(-24)}) \& 248 \& 136 \& 112 \& 4 \\ (E_8) \& (E_{8(+8)}) \& 248 \& 120 \& 128 \& 8 \\ \hline \end{tabular}$	(not rendered)	<table><tr><th>Root space</th><th>G'/H</th><th>Dim G'</th><th>Dim H</th><th>Dim P</th><th>Rank P</th></tr><tr><td>G_2</td><td>$G_{2(+2)}/(A_1\oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr><tr><td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr><tr><td></td><td>$F_{4(+4)}/(C_3\oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr><tr><td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr><tr><td></td><td>$E_{6(-14)}/(D_5\oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr><tr><td></td><td>$E_{6(+2)}/(A_5\oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr><tr><td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr><tr><td>E_7</td><td>$E_{7(-25)}/(E_6\oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr><tr><td></td><td>$E_{7(-5)}/(D_6\oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr><tr><td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr><tr><td>E_8</td><td>$E_{8(-24)}/(E_7\oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr><tr><td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr></table>	Root space	G'/H	Dim G'	Dim H	Dim P	Rank P	G_2	$G_{2(+2)}/(A_1\oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3\oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5\oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5\oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6\oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6\oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7\oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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gilmore-lie-groups_ TAB_026	240	■ 0.655	<pre>\begin{tabular}[t]{l l l l}\hline & Procedure & Result \\ \hline 1 & Locate singularities & $0, \infty$ \\ 2 & Determine analytic behavior at singular points & $r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$ \\ 3 & Keep only $\mathcal{L}^2$ solutions & $\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$ \\ 4 & Look for solutions with proper asymptotic behavior & $R = r^\gamma e^{\lambda r} f(r)$ \\ 5 & Construct differential equation for $f(r)$ & $\left[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$ \\ 6 & Construct recursion relation & $f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$ \\ 7 & Look at asymptotic behavior & $f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$) \\ 8 & Construct quantization condition & $2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$ \\ 9 & Construct explicit solutions & $E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$ \\ \hline \end{tabular}</pre>	(not rendered)	<table><thead><tr><th>Procedure</th><th>Result</th></tr></thead><tbody><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R = r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$\left[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$</td></tr></tbody></table>	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$\left[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$	6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)	8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$	9 Construct explicit solutions	$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$
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gilmore-lie-groups_TAB_027	244	■0.998	$\begin{tabular}{t l l l l } \hline \backslash \hline \backslash (m\backslash) \& \backslash (l=0\backslash) \& \backslash (l=1\backslash) \& \backslash (l=2\backslash) \& \backslash (l=3\backslash) \\ \hline 0 \& \backslash (\sqrt{\frac{1}{4\pi}}\backslash) \& \backslash (\sqrt{\frac{3}{4\pi}}\backslash) \& \backslash (\sqrt{\frac{5}{16\pi}}\backslash) \& \backslash (\sqrt{\frac{7}{16\pi}}\backslash) \\ \hline \pm 1 \& \backslash (\sqrt{\frac{3}{8\pi}}\backslash) \& \backslash (\sqrt{\frac{15}{8\pi}}\backslash) \& \backslash (\sqrt{\frac{21}{64\pi}}\backslash) \& \backslash (\sqrt{\frac{35}{64\pi}}\backslash) \\ \hline \pm 2 \& \backslash (\sqrt{\frac{3}{8\pi}}\backslash) \& \backslash (\sqrt{\frac{15}{32\pi}}\backslash) \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \\ \hline \pm 3 \& \backslash (\sqrt{\frac{3}{8\pi}}\backslash) \& \backslash (\sqrt{\frac{15}{32\pi}}\backslash) \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \\ \hline \end{tabular}$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$				
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± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$																																																							
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gilmore-lie-groups_TAB_028	258	■0.721	$\begin{tabular}{t l l l l } \hline H \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \\ \hline A_4 \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \\ \hline \Gamma_4 \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \\ \hline \Gamma_5 \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \\ \hline \Theta \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - 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gilmore-lie-groups_ TAB_033	290	■ 1.000	<pre>\begin{tabular}[t]{ l l l } \hline & Electric & Magnetic \\ \\ \hline Charge density & \(\rho_{\mathbf{x}}(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)\) & \(\rho_{\mathbf{x}}(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)\) \\ \\ \hline Current density & \(\mathbf{J}_{\mathbf{x}}(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}\) & \(\mathbf{J}_{\mathbf{x}}(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}\) \\ \\ \hline Conservation law & \(\nabla \cdot \mathbf{J}_{\mathbf{x}}(\mathbf{x}, t) + \frac{\partial \rho_{\mathbf{x}}(\mathbf{x}, t)}{\partial t} = 0\) & \(\nabla \cdot \mathbf{J}_{\mathbf{x}}(\mathbf{x}, t) + \frac{\partial \rho_{\mathbf{x}}(\mathbf{x}, t)}{\partial t} = 0\) \\ \\ \hline \end{tabular}</pre>	<table><tr><th></th><th>Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$	<table><tr><th colspan="2">Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>	Electric		Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$
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gilmore-lie-groups_TAB_034	311	0.852	$\begin{tabular}{ t { l l l l l l }} \hline \multicolumn{3}{ c }{\text{Infinitesimal generators}} \& \multicolumn{3}{ c }{\text{Canonical coordinates}} \& \text{Constraint} \\ \hline \xi(x,y) \& \eta(x,y) \& \zeta(x,y,p) \& R(x,y) \& S(x,y) \& T(x,y,p) \& dS/dR \\ \hline 0 \& 1 \& 0 \& x \& y \& p \& T \\ 1 \& 0 \& 0 \& y \& x \& p \& 1/T \\ 1/a \& -1/b \& 0 \& ax+by \& bx-ay \& p \& (b-aT)/(a+bT) \\ x \& 0 \& -p \& y \& \ln x \& xp \& 1/T \\ 0 \& y \& p \& x \& \ln y \& p/y \& T \\ x/a \& y/b \& (1/b-1/a)p \& y^b/x^a \& b \ln y \& p/x^{(a/b-1)} \& (bT/R)/(bT-aR^{(1/b)}) \\ x \& y \& 0 \& y/x \& \ln y \& p \& (T/R)/(T-R) \\ x \& -y \& -2p \& xy \& \ln y \& x^2p \& (T/R)/(T+R) \\ 2x \& y \& -p \& y^2/x \& \ln y \& yp \& (T/R)/(2T-R) \\ x \& 2y \& p \& y/x^2 \& \ln y \& p/x \& (T/R)/(T-2R) \\ y \& 0 \& -p^2 \& y \& x/y \& x-y/p \& -T/R^2 \\ 0 \& x \& 1 \& x \& y/x \& xp-y \& T/R^2 \\ -y \& x \& 1+p^2 \& \sqrt{x^2+y^2} \& \tan^{-1}(y/x) \& (y-xp)/(x+yp) \& -T/R \\ 1 \& y/x \& (px-y)/x^2 \& y/x \& x \& (xp-y)/x^2 \& 1/T \\ a \& x \& 1 \& x^2-2ay \& x/a \& x-ap \& 1/(2aT) \\ a \& y \& p \& x-a \ln y \& x/a \& p/y \& (1/a)/(1-aT) \\ x \& b \& -p \& e^y/x^b \& y/b \& e^y * p^b \& (bR)^{-1}/[1-b(R/T)^{(1/b)}] \\ \\ y \& b \& -p^2 \& y^2-2bx \& y/b \& y-b/p \& 1/(2bT) \\ 0 \& e^{f(x)} \& f'e^f \& x \& y/e^f \& p-yf' \& T/e^{f(R)} \\ x^2 \& xy \& y-xp \& y/x \& 1/x \& xp-y \& 1/T \\ xy \& y^2 \& yp-xp^2 \& y/x \& 1/y \& y/p-x \& 1/(TR^2) \\ xy \& 0 \& -yp-xp^2 \& y \& (\ln x)/y \& y/(xp)-\ln x \& T/R^2 \\ 0 \& xy \& y+xp \& x \& (\ln y)/x \& xp/y-\ln y \& T/R^2 \\ g(y) \& 0 \& -g'p^2 \& y \& x/g \& 1/p-xg'/g \& T/g(R) \\ 0 \& f(x) \& f' \& x \& y/f \& fp-f'y \& T/f^2(R) \\ f(x) \& 0 \& -f'p \& y \& F \text{ (} F'f=1 \text{)} \& pf \& 1/T \\ 0 \& g(y) \& g'p \& x \& G \text{ (} G'g=1 \text{)} \& p/g \& T \\ x^{k+1} \& kx^ky \& x^k(k^2y/x-p) \& y/x^k \& 1/x^k \& xp-ky \& -k/T \\ kxy^k \& y^{k+1} \& y^k(p-k^2xp^2/y) \& x/y^k \& 1/y^k \& y/p-kx \& -k/T \\ \hline \end{tabular}$	<table><tr><th colspan="2">Infinitesimal generators</th><th colspan="2">Canonical coordinates</th><th>Constraint</th></tr><tr><td>$\xi(x,y)$</td><td>$\eta(x,y)$</td><td>$R(x,y)$</td><td>$S(x,y)$</td><td>$T(x,y,p)$</td></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a \ln y$</td><td>x/a</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td></tr><tr><td>y</td><td>b</td><td>$-p^2$</td><td>y^2-2bx</td><td>y/b</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln 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y$	y	0	$-p^2$	y	x/y	0	x	1	x	y/x	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	1	y/x	$(px-y)/x^2$	y/x	x	a	x	1	x^2-2ay	x/a	a	y	p	$x-a \ln y$	x/a	x	b	$-p$	e^y/x^b	y/b	y	b	$-p^2$	y^2-2bx	y/b	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	x^2	xy	$y-xp$	y/x	$1/x$	xy	y^2	$yp-xp^2$	y/x	$1/y$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	0	xy	$y+xp$	x	$(\ln y)/x$	$g(y)$	0	$-g'p^2$	y	x/g	0	$f(x)$	f'	x	y/f	$f(x)$	0	$-f'p$	y	$F \text{ (} F'f=1 \text{)}$	0	$g(y)$	$g'p$	x	$G \text{ (} G'g=1 \text{)}$	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$
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gilmore-lie-groups_ TAB_035	316	■1.000	$\begin{tabular}[t]{ l l l }\hline 1830 & \text{Galois} & \text{solve algebraic equations} \\ \hline 1874 & \text{Lie} & \text{solve differential equations} \\ \hline 1969 & \text{Risch} & \text{integrate in closed form} \\ \hline ? & - & \text{solve determining equations for } \xi, \eta \\ \hline ? & - & \text{solve determining equations for } R, S, T \\ \hline ? & - & \text{solve } F(R, -, T) = 0 \text{ for } R. \\ \hline \end{tabular}$	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>-</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	-	solve determining equations for ξ, η	?	-	solve determining equations for R, S, T	?	-	solve $F(R, -, T) = 0$ for R .	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>-</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	-	solve determining equations for ξ, η	?	-	solve determining equations for R, S, T	?	-	solve $F(R, -, T) = 0$ for R .						
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gilmore-lie-groups_ TAB_036	318	■1.000	$\begin{tabular}[t]{ l l l }\hline \text{\(k\)} & \text{Potential type} & \text{Transformation} \\ \hline -1 & \text{Coulomb} & \gamma^2 = \beta^3 \\ \hline 0 & \text{no force} & \gamma^2 = \beta^2 \\ \hline +1 & \text{local gravitational potential} & \gamma^2 = \beta^1 \\ \hline +2 & \text{harmonic oscillator} & \gamma^2 = \beta^0 \\ \hline \end{tabular}$	<table><tr><td>k</td><td>Potential type</td><td>Transformation</td></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$	<table><tr><td>k</td><td>Potential type</td><td>Transformation</td></tr><tr><td>−1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	−1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$												
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